Physics 3181 — Spring 2023

Computational Physics

Physics 3181 Lectures: Mon/Wed 2:20-3:35 pm, Corcoran Hall, Room 413
Instructor: Prof. Oleg Kargaltsev
Office: Corcoran Hall 407; Office hours: Tuesday 3-4 pm, or by appointment
Email: kargaltsev@gwu.edu
Prerequisites: MATH 2233 and PHYS 3161, programming experience

Short description: The purpose of this course is to *discuss and practice* a number of numerical methods that are useful in solving Physics, Astrophysics, and Applied Mathematics problems. The course will cover a good fraction of methods that are frequently encountered in these fields. These methods are well established and have a solid theoretical foundation. Our focus will be less on the theoretical underpinnings and more on *practical use of these methods to solve concrete problems*.

Details: The course is project driven and a new topic will be discussed every week. To help you understand the material and to provide an assessment of your progress, you will have to solve a homework each week. The homework will be motivated by physics problems and to solve then you will rely on the numerical methods covered in class. Required physics background will be reviewed in class.

The programming tools we plan to use in this course are Python (or *C) and Wolfram Mathematica*. We will use Python (or C) to practice low-level programming and solve some basic computational problems. Mathematica will be used for higher-level problems, to assist with debugging the codes and verifying Python (C) code results, to further analyze and visualize the results of computations, to create plots and animations, etc. For each homework set, you will be asked to submit the documented code and, in some cases, write a report (preferably in Latex, a tool that will allow you to typeset equations, include graphs to support your presentation, and create a bibliography). An effective workflow depends on your ability to use these tools. Although some introduction will be provided, this course is not meant to teach you programming language or Latex, so you will need to work on it outside the class. Students must bring their own laptops and configure them appropriately. The use of Unix (the one in Mac OS or other flavor) is preferred (your instructor will have laptop with Mac OS).

I currently prefer using Python over C because Python use has become much more widespread in physics and astronomy although in certain areas of physics C is still the language of choice.

If you are not familiar with Python, I can recommend a brief introduction that can be found here: https://blogs.umass.edu/candela/computational-physics-in-python/?utm_source=pocket_reader . Please review and try before our course begins. Please install Python 3 and JupyterLab (unless you prefer other Python IDE like Spyder). If you are not very good at installing things it is probably best to install via Anaconda (see https://runestone.academy/ns/books/published/httlads/PythonReview/anaconda.html). Please also install Wolfram Mathematica (available for free from GW) before you come to class.

Learning Outcomes: Upon completion of this course you will be able to write programs in Python (or C) and Mathematica and apply programming skills in conjunction with the laws of physics to

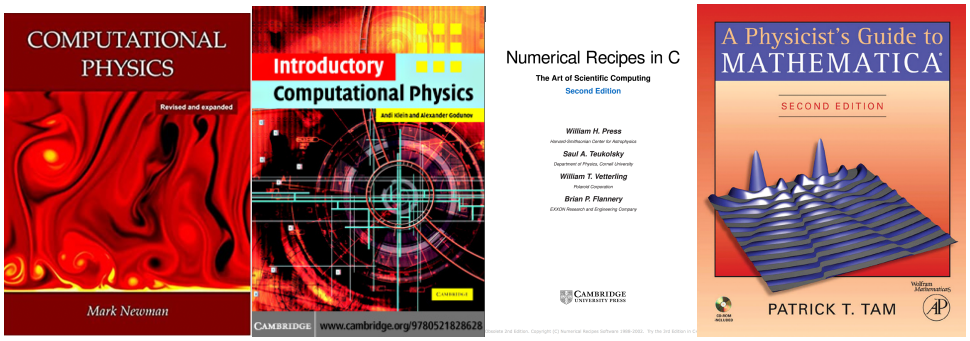
solve a wide range of physics and astrophysics problems .

Students will acquire the following *specific skills* that will be verified via specially designed, graded homework assignments:

- Numerical Integration
- Integration of differential equations
- Optimization
- Hypothesis testing and fitting
- Chaos/N-body dynamics
- Machine Learning
- Monte Carlo methods
- Root finding
- Fourier transforms

Textbook: No textbook is required. The lecture slides and notes will be provided by the instructor as needed. They will be accessible in GW Blackboard.

If you still want to have a textbook, I can recommend:



Mark Newman "Computational Physics" (ISBN-13: 978-1480145511)

Klein, Cambridge University Press 2006, "Introductory Computational Physics"

Tam, Cambridge University Press 2006 "A Physicist's Guide to Mathematica"

Numerical Recipes in C available free at http://www.nrbook.com/a/bookcpdf.html

If you want to purchase a book and only a single one, I recommend Python-based Mark Newman's "Computational Physics" (ISBN-13: 978-1480145511).

Assignments: Coding homework projects will be posted regularly at the class web site on Blackboard http://blackboard.gwu.edu. Homework must be turned in as a printed report. The report must contain statement of the problem, algorithm/method description, code, results of calculations in numerical and/or graphical form, physical interpretation of the results by student. The assignments will be graded by the instructor using rubric assessment for each of these components. Assignments may also require turning is a documented code in Python (or C) or/and Mathematica (uploaded to Blackboard or sent by e-mail). Students must visit the class Blackboard site regularly.

Expected Time Investment: Over 14 weeks students will spend 3 hours per week in classroom activities (lectures, quizzes, problem-solving and programming projects). Minimum amount of independent, out-of-class, learning expected per week is 8 hrs.

Outline of the course for Spring 2023

Week	Dates	Topic
1	Jan 18	Introduction and setup. Basics of programming (in Python/C).
2	Jan 23,25	Basics of Mathematica. Graphing in Mathematica. Data visualization. Intro to statistics. Predator and prey problem.
3	Jan 30, Feb 1	Roots of nonlinear equations. Interpolation, cross-correlation, numerical derivative. KS-test. Sums. Multiple collision accelerator. Resistor networks.
4	Feb 6,8	Numerical integration. Random number generation.
5	Feb 13,15	Ordinary differential equations I. Euler method and advanced difference schemes for ODEs. Nonlinear pendulum. Intro to Chaos. Motion in central field.
6	Feb 22	Ordinary differential equations II. Boundary value problems for ODEs. Linear algebra. Quantum particle in a box.
7	Feb 27, Mar 1	Basics of fitting and minimization. Chi-square fitting (X-ray spectrum example).
8	Mar 6,8	Bootstrap and Monte Carlo methods. Applications to fitting and image analysis. Mid semester review.
9	Mar 13,15	**Spring break!**
Spring break	Mar 20,22	N-body simulations (star cluster, gas dynamics). Increasing computational efficiency and power. High-performance computing.
10	Mar 27, 29	Unsupervised Machine Learning. Clustering (k-NN algorithm and its relatives)
11	Apr 3, 5	Supervised Machine Learning. Information Gain. Entropy. Decision Trees.
12	Apr 10,12	Time variable signals. Detecting variability.
13	Apr 17,19	Periodicity detection and search for periodic signal. Fourier and its applications to signal processing.
14	Apr 24,26	Term project discussion. Review.
15	May 1	Data visualization.
16	May 8	Term project presentations

Attendance: The students are *expected to attend every class*. If you have to miss >2 classes, you should discuss this with the instructor. Note that if you miss a class, your grade for the in-class quiz/participation will be zero. All requests to have an absence excused must be substantiated with appropriate documentation. Valid excuses include personal illness (with a doctor's note), family emergency, GW sanctioned travel for a team sport, or religious holiday.

Course Grade: The final grade will be based on the homework (60%), term project (20%), and attendance, in-class participation, and quizzes (20%). There is no final exam in this course.

ONLINE RESOURCES

The course webpage will be on Blackboard (http://blackboard.gwu.edu/). Course announcements will be posted on this site. Any distributed reading materials and any PowerPoint slides shown in class will be made available online *after* each class. After entering *Blackboard*, it is necessary for you to click on the course name (PHYS 3181). You are *automatically* subscribed within the *Blackboard* system to the courses for which you are registered (but you also must have a valid GW e-mail address to login).

Numerical to letter grade conversion

≥ 94.00	**A**	70.00 – 73.99	**C**
90.00 – 93.99	**A-**	66.00 – 69.99	**C-**
86.00 – 89.99	**B+**	61.00 – 65.99	**D+**
82.00 – 85.99	**B**	55.00 – 60.99	**D**
78.00 – 81.99	**B-**	50.00 – 54.99	**D-**
74.00-77.99	**C+**	≤ 49.99	**F**

University policies

Use of Electronic Course Materials and Class Recordings

Students are encouraged to use electronic course materials, including recorded class sessions, for private personal use in connection with their academic program of study. Electronic course materials and recorded class sessions should not be shared or used for non-course related purposes unless express permission has been granted by the instructor. Students who impermissibly share any electronic course materials are subject to discipline under the Student Code of Conduct. Please contact the instructor if you have questions regarding what constitutes permissible or impermissible use of electronic course materials and/or recorded class sessions. Please contact Disability Support Services at disabilitysupport.gwu.edu if you have questions or need assistance in accessing electronic course materials.

Academic Integrity Code

Academic Integrity is an integral part of the educational process, and GW takes these matters very seriously. Violations of academic integrity occur when students fail to cite research sources properly, engage in unauthorized collaboration, falsify data, and in other ways outlined in the Code of Academic Integrity. Students accused of academic integrity violations should contact the Office of Academic Integrity to learn more about their rights and options in the process. Outcomes can range

from failure of assignment to expulsion from the University, including a transcript notation. The Office of Academic Integrity maintains a permanent record of the violation.

More information is available from the Office of Academic Integrity at studentconduct.gwu.edu/academic-integrity. The University's "Guide of Academic Integrity in Online Learning Environments" is available at studentconduct.gwu.edu/guide-academic-integrity-online-learning-environments. Contact information: rights@gwu.edu or 202-994-6757.

University policy on observance of religious holidays

In accordance with University policy, students should notify faculty during the first week of the semester of their intention to be absent from class on their day(s) of religious observance. For details and policy, see "Religious Holidays" at provost.gwu.edu/policies-procedures-and-guidelines.

Support for students outside the classroom

Virtual academic support

A full range of academic support is offered virtually in the fall 2020 and spring 2021 semesters. See coronavirus.gwu.edu/top-faqs for updates.

Tutoring and course review sessions are offered through Academic Commons in an online format. See academiccommons.gwu.edu/tutoring

Writing and research consultations are available online. See academiccommons.gwu.edu/writing-research-help

Coaching, offered through the Office of Student Success, is available in a virtual format. See studentsuccess.gwu.edu/academic-program-support

Writing Center

GW's Writing Center cultivates confident writers in the University community by facilitating collaborative, critical, and inclusive conversations at all stages of the writing process. Working alongside peer mentors, writers develop strategies to write independently in

academic and public settings. Appointments can be booked online. See gwu.mywconline.

Academic Commons

Academic Commons provides tutoring and other academic support resources to students in many courses. Students can schedule virtual one-on-one appointments or attend virtual drop-in sessions. Students may schedule an appointment, review the tutoring schedule, access other academic support resources, or obtain assistance at academiccommons.gwu.edu. Academic Commons offers several short videos addressing different virtual learning strategies for the unique circumstances of the fall 2020 and spring 2021 semesters. See academiccommons.gwu.edu/study-skills. They also offer a variety of live virtual workshops to equip students with the tools they need to succeed in a virtual environment. See tinyurl.com/gw-virtual-learning

Disability Support Services (DSS) 202-994-8250

Any student who may need an accommodation based on the potential impact of a disability should contact Disability Support Services to establish eligibility and to coordinate reasonable accommodations. disabilitysupport.gwu.edu

Counseling and Psychological Services 202-994-5300

GW's Colonial Health Center offers counseling and psychological services, supporting mental health and personal development by collaborating directly with students to overcome challenges and difficulties that may interfere with academic, emotional, and personal success. healthcenter.gwu.edu/counseling-and-psychological-services

GW Title IX statement:

The George Washington University (GW) and its faculty are committed to creating a safe and open learning environment for all students. If you or someone you know has experienced sexual harassment, including sexual assault, dating or domestic violence, and stalking, please know that help and support are available. GW strongly encourages all members of the community to take action, seek support, and report incidents of sexual harassment to the Title IX Office. You may contact the Title IX Office at 202-994-7434 or at titleix@gwu.edu or learn more by visiting titleix.gwu.edu. Please be aware that faculty members are required to disclose information about suspected or alleged sexual harassment or other potential violations of the Title IX Sexual Harassment and Related Conduct Policy to the Title IX Office. If the Title IX Office receives information about an incident, they will reach out to offer information about resources, rights, and procedural options as a member of the campus community. Community members are not required to respond to this outreach. If you, or another student you know, wishes to speak to a confidential resource who does not have this reporting responsibility, please contact Counseling and Psychological Services through the Colonial Health Center 24/7 at 202-994-5300, or the Office Of Advocacy and Support at 202-994-0443 or at oas@gwu.edu.